



Solution Guide: Creating Aliases

In this exercise, you created custom commands using aliases and the `~/ .bashrc` file.

- You had to create the following aliases:
 - A custom `ls` command.
 - Custom commands to change directories into `Documents`, `Downloads`, and the `/etc` directory.
 - A custom command to easily edit the `~/ .bashrc` file.
 - Custom commands for some of the compound commands you created in the previous activity.
 - You also had to reload the `.bashrc` file so that the commands took effect.
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Solution

Log into the lab environment with the username `sysadmin` and password `cybersecurity`.

You can either add commands directly inside `~/ .bashrc` using `nano`, or you can use output redirection to append them to the `~/ .bashrc` file.

Important: Remember, you must use `>>` and not `>`, or else you will overwrite the entire file. It's recommended to make a backup of the `~/ .bashrc` file before changing it.

Start by creating a backup copy of your `~/ .bashrc` file by running `cp ~/ .bashrc ~/ .bashrc.bak`.

1. Create an alias in your `~/ .bashrc` file for `ls -a`.
 - Type `echo "alias lsa='ls -a'" >> ~/ .bashrc`.
 - `alias` indicated that the following code is an alias.
 - `lsa=` is the name of the new command. We can use anything we want, but we want to be careful not to use a command that already exists.
 - Note that we have to use a mixture of double and single quotes (`"` and `'`) here to get this command to work correctly.
 - `'ls -a'` is the command we are creating the alias for.
 - The `echo` command is wrapped in double quotes (`"`) and the alias is wrapped in single quotes (`'`).

- `alias lsa='ls -a'` is the only line we need to add to our `~/.bashrc` file. If we wanted to add this directly to the `bashrc` file, we could use `echo` and redirection to do it in one line.
- Run `echo "alias lsa='ls -a'" >> ~/.bashrc`.
 - We could chain it together with `&& source ~/.bashrc` to automatically reload the file and enable the new alias.
 - Example: `echo "alias lsa='ls -a'" >> ~/.bashrc && source ~/.bashrc`
- 2. Create an alias in your `~/.bashrc` file for `cd ~/Documents`, `cd ~/Downloads`, and `cd /etc`.
Use the following command structure `alias docs='cd ~/Documents'` for each directory.
 - `~/Documents`:
 - Run `echo "alias docs='cd ~/Documents'" >> ~/.bashrc`
 - `~/Downloads`:
 - Run `echo "alias dwn='cd ~/Downloads'" >> ~/.bashrc`
 - `~/etc`:
 - Run `echo "alias etc='cd /etc'" >> ~/.bashrc`

These are the only lines needed for the `~/.bashrc` file.

Take a moment to see what's happening to the `~/.bashrc` file.

- Run `tail -4 ~/.bashrc`
You should get output similar to:

Unset

```
alias lsa='ls -a'

alias docs='cd ~/Documents'

alias dwn='cd ~/Downloads'

alias etc='cd /etc'
```

Bonus Aliases

Create aliases for the following:

- `nano ~/.bashrc`

- `mkdir ~/research && cp /var/log/* /etc/passwd /etc/shadow /etc/hosts ~/research`

Create an alias in your `~/ .bashrc` file for `nano ~/ .bashrc`.

- Run `echo "alias rc='nano ~/ .bashrc'" >> ~/ .bashrc`.
- Run `source ~/ .bashrc` to reload the file and enable our commands.
- Run `lsa` to demonstrate your custom `ls` command.
- Run `docs` to demonstrate your custom `cd` command.
- Run `rc` to demonstrate your custom `nano ~/ .bashrc` command.

Scroll to the bottom where the aliases are being added.

- The section should look like:

Unset

```
alias lsa='ls -a'

alias docs='cd ~/Documents'

alias dwn='cd ~/Downloads'

alias etc='cd /etc'

alias rc='nano ~/ .bashrc'
```

Add a comment above your aliases to mark the section:

Unset

```
# Custom Aliases

alias lsa='ls -a'

alias docs='cd ~/Documents'

alias dwn='cd ~/Downloads'

alias etc='cd /etc'

alias rc='nano ~/ .bashrc'
```

Complete the same steps for the following:

1. `mkdir ~/research && cp /var/log/* /etc/passwd /etc/shadow /etc/hosts ~/research`
 - o **Solution:** `echo "alias logs='mkdir ~/research && cp /var/log/* /etc/passwd /etc/shadow /etc/hosts ~/research'" >> ~/.bashrc`

The Custom Aliases section should now look like:

Unset

```
# Custom Aliases

alias lsa='ls -a'

alias docs='cd ~/Documents'

alias dwn='cd ~/Downloads'

alias etc='cd /etc'

alias rc='nano ~/.bashrc'

alias logs='mkdir ~/research && cp /var/log/*
/etc/passwd /etc/shadow /etc/hosts ~/research'
```

We can either keep the output file redirection `>> ~/research/users.txt` or we can leave it out. If we do leave it out, we can still use redirection when we run our custom alias.

- Save and quit Nano.
- Type `exec >> ~/research/users.txt` as an example of using redirection with a custom alias.

Remember, with every edit, you will need to reload the `~/.bashrc` file before the edits will take effect.

- Run `source ~/.bashrc`.
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